Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/21/2017
PI Name:	Feinberg, Andrew M.D.		
Project Title:	Comprehensive Whole Genome Analysis of Differential Epigenetic Effects of Space Travel on Monozygotic Twins		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Baltimore	**State:**	MD
Zip Code:	21205-1503	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/14/2014	**End Date:**	04/13/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 4/30/2018 per NSSC information (Ed., 3/21/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14AH28G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Several substantial medical barriers to long-term space travel include radiation damage, weightlessness, and stress caused by the specialized spacecraft environment. This Human Exploration Research Opportunities (HERO) grant solicitation requests applicants to take advantage of the unusual opportunity to compare identical twins, one of whom will be traveling in orbit for a year while the other will remain on Earth. The area of science that most directly addresses this opportunity is epigenetics, the study of chemical changes to DNA and the associated proteins, other than the DNA sequence itself. Our laboratory is the leading center that has pioneered the methods for analyzing epigenetics at a genome-scale level, termed epigenomics, and for applying these methods to the discovery of epigenetic causes and mediators of common human disease. These discoveries include the first analysis of epigenomic changes in individuals over time, the first complete epigenomic map of human cancer, the discovery of an epigenomic basis for autoimmune disease, and epigenomic mechanisms for injury response. In this proposal, we will test the hypothesis that space-induced exposure causes epigenome damage. We will determine the nature of this damage, and whether it is correctible spontaneously. We will measure three epigenome-related modalities in identical twins, namely: DNA methylation, a chemical change in the base cytosine; large heterochromatin regions related to chemical modifications of histones; and their functional relationship to gene expression of RNA (expression data generated by other Twins' investigators). We will obtain measurements prior to launch, after reentry, and at quarterly intervals in space, as well as after any unforeseen environmental exposure such as a spike in radiation. We will complement this analysis with whole genome DNA sequencing data generated by other Twins' researchers before and after the flight on both twins, in order to relate epigenomic damage to subsequent DNA damage. Given the potentially reversible nature of epigenetic changes, this work may help to identify or mitigate exposure to radiation, weightlessness, and environmental stressors, which are considered the greatest barriers to long term space flight. Thus our work could have a huge impact on long-duration space travel, if we can identify potentially reversible epigenetic precursors of genetic damage to astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The Principal Investigator and his staff have spent considerable effort optimizing protocols for cell fractionation, DNA and chromatin analysis, and sequencing. We have successfully obtained adequate yields on test samples for analysis of fractionated PBMC, including CD4+, CD8+, and CD19+ fractions. To date, we have generated whole genome bisulfite sequencing (WGBS) libraries for all samples received and have completed sequencing of CD4+ cells from 4 time points in each twin, to measure DNA methylation at a genome-wide scale. Novel mathematical modeling tools we recently developed allow us to examine paired samples to identify regions of the genome where the epigenome differs in a meaningful way. Using these methods, we will interrogate our data for epigenetic changes that arose during Scott Kelly's time in space and returned (or not) to baseline levels upon his return. We will then compare these changes to the same regions in Mark Kelly to see if they remained unchanged or were altered in a different way. Preliminary results are summarized below: • CD4+ samples analyzed via WGBS: o Mark – pre-flight (12/3/14), in-flight (7/30/15 and 1/20/16), post-flight (7/17/16) o Scott – pre-flight (10/16/14), in-flight (6/11/15 and 12/12/15), post-flight (6/14/16) • We found subtle, but global differences in genome-wide DNA methylation: o Methylation very similar in twins pre-flight. o In general, Scott's DNA methylation is decreased in-flight and returns to near baseline upon return. o Mark's DNA methylation level initially decreased (similar to Scott), but increased at the second in-flight time point before returning to near baseline. We cannot ascribe clinical significance to these changes in DNA methylation or relate to a known human health effect as DNA methylation is not a CLIA-approved diagnostic assay. However, these changes could help in understanding the effects of future space flight by indicating which genes or regions of the genome are most responsive to the space environment. Our data analysis methods are somewhat unique in that we are also able to assess biological noise, or "entropy" in the epigenome, a measure of lability in gene activity. We found changes in entropy between the twins, as well. While again not of known clinical significance, these results could indicate genes that are more sensitive to a changing environment, whether on Earth or in space. We are coordinating with data managers with Mike Snyder's group to handle all data uploads so that all Twins' data can be easily accessed by all the Twins' researchers.
Bibliography Type:	Description: (Last Updated: 05/20/2019)
Significant Media Coverage	Witze A. "Astronaut twin study hints at stress of space travel. Article about NASA's TWINS study findings, quoting PI Feinberg." Nature News. January 26, 2017. http://dx.doi.org/10.1038/nature.2017.21380 , Jan-2017
Significant Media Coverage	Khan A. "What are the long-term health effects of living in space? NASA is studying twins Mark and Scott Kelly to find out. Article quotes PI and includes information about PI's research as part of NASA's Twins Study." Los Angeles Times newspaper, September 7, 2016. http://www.latimes.com/science/la-sci-nasa-twin-study-20160707-snap-story.html , Sep-2016